

Mobile Game A/B Testing Project

A plain-English walkthrough for someone new to Product Analytics

1. What does a Product Analyst actually do (in short)

You already know Data Analyst / Business Analyst work - pulling data, building dashboards, finding insights for the business. Product Analyst is close to that, but the main focus is different:

- A Product Analyst mostly answers questions about a **product's** behaviour - do people use this feature, do they come back, does a change to the product help or hurt usage.
- The main tool of the trade is the **A/B test** (also called an experiment) - showing two versions of something to two different groups of users and comparing the outcome.
- The core skill is being able to say, with actual statistical proof, whether a change caused a result - not just "it looks like it went up".

This project is a classic PA case study: a game company wants to test a product change (where a paywall/gate sits) and needs someone to tell them, with data, whether to ship it or not.

2. The business problem, in one paragraph

Cookie Cats is a mobile puzzle game. At level 30, the game stops the player and makes them either wait a while or pay to keep playing - this is called a **gate**. The product team has an idea: what if we move that gate to level 40 instead, so players get more free levels before hitting the paywall? Will that make people stick around longer (retention), or will it backfire?

Think of it like a free trial. Should the paywall come earlier or later? Earlier might annoy people faster, but later might make them binge and get bored before they even hit it. Nobody knows for sure - that's why you test it instead of guessing.

3. The data - what you're working with

90,188 real users who installed the game during the test period. Each user was randomly placed into one of two groups:

- **gate_30** (control group) - the game as it currently is, gate at level 30. ~44,700 users.
- **gate_40** (treatment group) - the new version being tested, gate at level 40. ~45,489 users.

For each user you also have:

- **sum_gamerounds** - how many rounds they played in their first 14 days (this tells you how engaged/active they are)
- **retention_1** - True/False, did they open the game again 1 day after installing

- **retention_7** - True/False, did they open the game again 7 days after installing

Retention is the metric that matters here. If people stop opening the app, the game eventually dies, no matter how good day-1 numbers look. retention_7 (one full week later) is generally seen as a much stronger signal of whether someone actually likes the game, versus retention_1 which just shows initial curiosity.

4. What was actually done, step by step

Step 1 - Clean the data first

Before touching any statistics, always look for junk in the data. Checked for missing values (none), duplicate users (none), and then looked at the distribution of sum_gamerounds.

Found one single user with almost 50,000 rounds played in 14 days. That's not humanly possible (it works out to playing non-stop, day and night, for two weeks straight). This is almost certainly a bot or test account, so it was removed before doing any analysis - one weird outlier like that can quietly distort an average or a model.

Interview tip: this is a very commonly asked thing - "how did you handle outliers/data quality issues?" You have a real, concrete answer here, use it.

Step 2 - Check the groups are actually random and balanced

Before comparing gate_30 vs gate_40, you want to make sure the split itself wasn't biased in some way (e.g. one group accidentally getting way more users, or one type of user). Ran a Chi-Square test on the group sizes - came back fine, no sign of a broken random split. This is just a sanity check step, standard practice before trusting any A/B test result.

Step 3 - Look at the raw retention numbers

Simplest step: just compare the average retention rate of each group.

Metric	gate_30 (control)	gate_40 (treatment)
Retention Day 1	44.8%	44.2%
Retention Day 7	19.0%	18.2%

Both numbers are slightly lower for gate_40. But a small difference in a sample could just be random luck. That's exactly why you don't stop here - you need a statistical test to know if the gap is **real** or just **noise**.

Step 4 - Hypothesis testing (the core of A/B testing)

This is the main concept interviewers will probe. In plain words:

- **Null hypothesis (H0):** "There is no real difference between gate_30 and gate_40 - any gap we see is just random chance."
- **Alternative hypothesis (H1):** "There IS a real difference."
- You run a statistical test (here, a **Two-Proportion Z-Test**, made for comparing two percentages) which gives you a **p-value**.

- **P-value in plain English:** if H0 were actually true (no real difference), the p-value is the chance you'd still see a gap this big just by random luck. A small p-value (usually under 0.05) means the gap is unlikely to be pure luck, so you reject H0 and conclude the difference is real.

Test	Result	What it means
Retention Day 1	p = 0.074	Not significant - can't confidently say there's a real difference
Retention Day 7	p = 0.0016	Significant - very unlikely to be random chance, gate_30 really is better

So: Day-1 retention gap could just be noise, but the Day-7 gap is real, and it favours keeping the gate at level 30.

Step 5 - Bootstrap (double-checking the result)

A bootstrap is a way of stress-testing your result without needing to trust a textbook formula. Instead of one test, the computer:

- Takes the 90,188 users, and randomly resamples the same number of users **with replacement** (so some users get picked more than once, some not at all) - this creates one "fake but plausible" version of the dataset
- Recalculates the retention_7 gap between gate_30 and gate_40 on that resample
- Repeats this 5,000 times, building 5,000 slightly different versions of the answer

If the result is real, almost all 5,000 versions should agree on the direction of the gap. Here, **99.98% of the 5,000 resamples** showed gate_30 ahead of gate_40 - basically total agreement. This backs up the z-test and makes the finding much harder to argue with in a meeting.

Step 6 - Cohort analysis by engagement level

A **cohort** just means a group of users who share something in common, that you track together. Normally in an app you'd group people by signup week ("the July cohort"), but this dataset has no dates - so instead, users were grouped into 4 cohorts based on how much they already play (Low, Mid-Low, Mid-High, High, based on sum_gamerounds).

Then retention_7 was compared across gate_30 vs gate_40, **within each cohort separately**, instead of just looking at one overall average. This is a very "product analyst" way of thinking - averages can hide the real story, so you break the population into meaningful groups and check each one.

Finding: the gap between gate_30 and gate_40 is small for casual players (Low/Mid-Low cohorts), but much bigger for the more engaged players (Mid-High/High cohorts). In other words, your best, most engaged players are the ones hurt most by moving the gate to level 40.

Step 7 - ANOVA (checking if that cohort difference is statistically real)

ANOVA (Analysis of Variance) is just a test that tells you if the differences you see across multiple groups are statistically meaningful, or could just be random spread. Here it was used to check something specific: does the version effect (gate_30 vs gate_40) behave differently depending on the cohort? This is called an **interaction effect**.

Result: yes, the interaction is statistically significant. This confirms the cohort chart wasn't just a coincidence-looking chart - the gate change genuinely affects engaged and casual players differently.

Step 8 - Logistic regression (one last check)

Logistic regression is used here to predict a Yes/No outcome (retained or not) using multiple factors at once, instead of one at a time. It answers: "If I account for how much someone already plays, does the gate version still matter on its own?"

Answer: yes. Even after controlling for engagement level, being in gate_40 still independently lowers the chance of retention, and this was statistically significant ($p < 0.001$). So it's not just that gate_40 happened to get less engaged users by chance - the gate placement itself has a real, independent effect.

5. The final answer - what you'd tell the product team

- Don't move the gate to level 40.
- Day-1 retention barely changes, and Day-7 retention gets meaningfully worse under gate_40.
- The drop is not spread evenly - it hits your most engaged players hardest, and those are usually your highest-value users (most likely to spend money or watch ads).
- If the team still wants to explore a later gate, test it only on high-engagement users first, rather than rolling it out to everyone.

6. Quick glossary - if an interviewer uses a term you blank on

A/B Test - Showing 2 versions of something to 2 random groups and comparing an outcome, to see if a change actually causes a difference.

Control vs Treatment - Control = the current/existing version. Treatment = the new version being tested.

Retention - Whether a user comes back and uses the product again after some time period (day 1, day 7, etc).

Hypothesis Test - A structured way to check if a difference you observe is real or just random chance.

P-value - The probability of seeing a result this extreme if there was actually no real effect. Smaller = more confident the effect is real. Usual cutoff: 0.05.

Z-test (two-proportion) - A specific hypothesis test used when comparing two percentages/rates (like two retention rates).

Bootstrap - Resampling your data thousands of times to see how stable/repeatable your result is, without relying purely on a formula.

Cohort - A group of users who share something in common (signup date, engagement level, etc.) that you analyze as a unit.

ANOVA - A test to check if differences across 3+ groups (or an interaction between two factors) are statistically real.

Interaction effect - When the effect of one thing (e.g. gate version) depends on another thing (e.g. how engaged the user is), rather than being the same for everyone.

Logistic Regression - A model used to predict a Yes/No outcome using multiple input variables at once.

7. Likely interview questions on this project

Q: Why did you use a Z-test and not a T-test?

A: Z-test is used when comparing two proportions/percentages (like retention rate), which is a binary yes/no outcome averaged over a large sample. T-test is more for comparing two averages of continuous numbers (like average revenue).

Q: Why bootstrap if you already had a p-value?

A: The z-test relies on some mathematical assumptions about the data's distribution. Bootstrapping doesn't assume anything, it just directly resamples the real data thousands of times, so it's a good way to double check the p-value result actually holds up.

Q: What would you have done if Day-1 AND Day-7 were both not significant?

A: I'd have told the team the test didn't show enough evidence either way, and either recommended not shipping the change (since there's no proven upside) or extending the test to collect more users/more time before deciding.

Q: How did you make sure the two groups were comparable?

A: Checked group sizes were close to 50/50 with a Chi-Square test, and made sure there was no major imbalance that would point to a broken random assignment.

Q: Why look at cohorts instead of just the overall average?

A: Because an overall average can hide the real story - here, casual players barely reacted to the gate change but engaged players reacted a lot. If you only look at the average, you'd underestimate the risk to your best users.

Q: What's your actual recommendation and why?

A: Keep the gate at level 30. The data shows Day-7 retention is significantly better there, and the drop under gate_40 is concentrated in the more engaged, likely higher-value users - which is the group you least want to lose.

Tip: read through this once, then try explaining Steps 4-8 out loud in your own words without looking. That's usually enough to sound confident in an interview.